The Quick and Dirty Guide

ArcGIS 10 and Python

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Advanced Settings, Docking and Search

- C:\Program Files\ArcGIS\Desktop10.0\Utilities\AdvancedArcMapSettings.exe
  - Change the temp folder
- Docking Workspaces
  - Allows you to customize the desktop by dragging the tools
  - CAUTION defaults to project workspace in your Documents > ArcGIS or My Documents > ArcGIS folder **reset to scratch drive.**
- Search – tab in TOC (our use is limited as it needs set up with ArcGIS server)

Catalog

- Integrated into ArcMap
- Use instead of Add Data as it will be quicker to find and drag drop (takes getting used to)
- Use to access tools/models from instead of opening another frameset.

TOC

- TOC – not much change
  - Visual changes
  - Tools menu is now Customize
  - Commands for geocoding, adding XY data, and adding linear referencing now File pull-down menu in a new pull-right menu called Add Data
  - Graphs and reports now in View pull-down menu

Geoprocessing

- New primary toolbar item.
  - Right-click your custom model or script tool and click Properties. On the General tab, uncheck Always run in foreground. A sand-timer icon displays while a process is running and you can click on Messages to see what is happening in real time (depends on the tool code).
  - Note that if the tool is using a layer in the current map it will ALWAYS run in the foreground to prevent duplicate editing.
  - Can add frequently used tools to this menu
  - The results view allows one to check on progress and rerun tools with different inputs but maintain settings or change settings but have same inputs etc without having to set all parameters again.
  - Model builder and Environment settings are accessed from here.

Base Maps

- Add from WMS services (openstreetmap is ok, others have too low resolution)
- Could be useful if we can connect into the new TRC map server
- Current use is mostly to speed up redraw rates
- Place all data that is relatively static ex. DCDB, roads, hydrology etc into a properly symbolized “Group Layer” called a “Base Map Layer” that would be a base for the map.
  - We can create a layer package for this and users can then add this to all new maps, it will automatically use the new DCDB etc.
- Only for vector – for Rasters there is Accelerated raster rendering

Editing

- Right click access to editing
- Define the edit type by creating templates.
- Easy access with mini-toolbars
- Snapping integrated into view environment as well (Space bar will suspend or enable snapping).
- Annotation is built into this
- Attachments
- A pop-up toolbar appears when editing (feature construction), this can be moved by hitting Tab

Selection Tools

- 4 new selections
  - Polygon
  - Lasso
  - Circle
  - Line

  They are based on the snapping environment.

Tables

- Mutiple tables in excel like tabs (custom arrangement possible) -
- Validate joins/relates before processing for error correction

LYR

- Can now be created for multiple files (not just one)
- Layer dataset (copy of all files to send to people)
- Layer group (link to all files with current custom properties) –must be in a group (r-click)
- Can be opened in 9.3.1
- Use representation to define symbology that will be used for the layer.

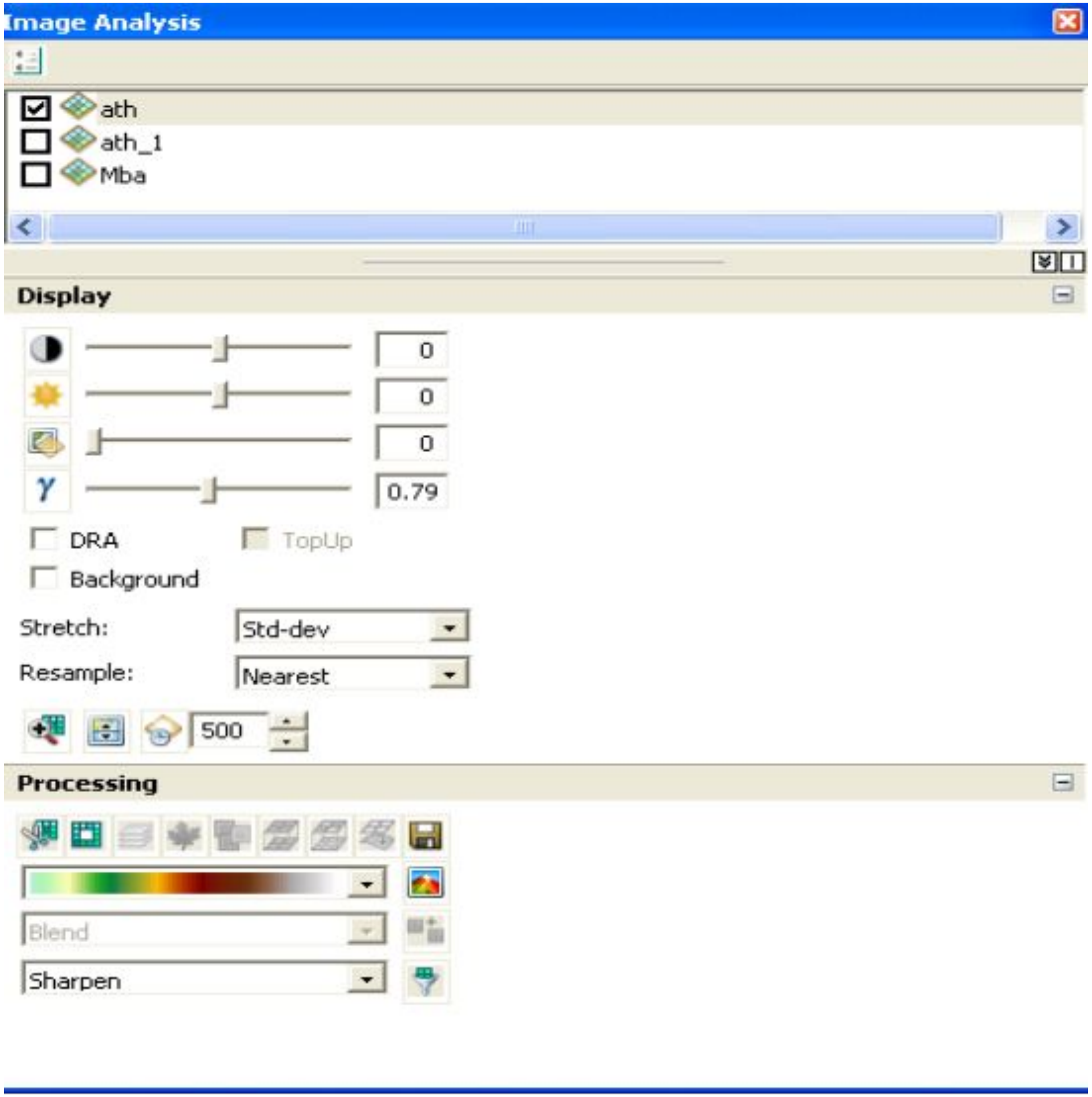
Field Calculator

- No more VBA code
- Python and VB Script with wizard

Rasters

- Pyramids are now .ovr files (compressible)
- On the fly mosaic (Mosaic Dataset) similar in MapInfo 10.5 –
- New tools to process these (windows->Image Analysis)
- New icons to represent diff. types
- Accelerated raster rendering

windows□Image Analysis



Geo-Databases

- For efficient use of the tools our data needs to be processed within FGDB's.
  - Don't use PGDB's

Model Builder

- "undo" or "redo" button
- user can include tools and models in the toolbar
- dock these tools on the toolbar
- Access from Catalog without having a toolbar using screen space.

Misc

- Map Packages
- Temporal data
- Animation
- Graphing and reporting
- Symbols –search without knowing name
- CAD -Improved support of different types

- maybe useful to show population growth/town growth over time for council meetings
- maybe to show reduction of forest cover, change in land use etc

Data Driven Pages

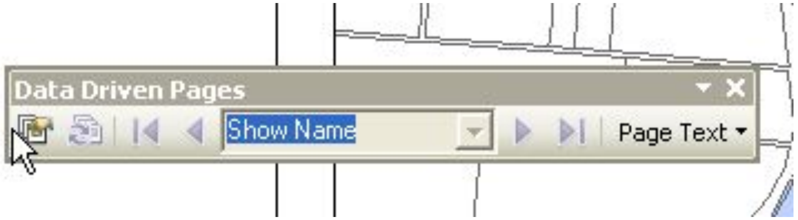
- Dynamic elements
  - Geographic extent of the map
  - Map Scale
  - North Arrow (if map rotation is set)
  - Scale bar
  - Scale text
- Dynamic text (for example, page name and page number)
  - Layers containing a dynamic query
  - Static elements
  - Size and orientation of layout page
  - Size and position of data frame(s)
  - Static text
  - Neatlines
- Exporting/Printing
  - Individual graphic/pdf files
  - Incorporate into a multipage PDF. .

Live Example - DDP

- Different methods to define area of each map
  - arcGIS
    - Create index grids along polyline (not polygon.) *There are issues when the dataset is large and loops back close to itself.*
    - create index grid within polygon area
  - Custom
    - create shp file of extents of all dataframes in existing MXD’s or Rasters
  - other advanced tools to define the adjacent maps, the angle, central point etc.
- The layer you want to use as the Index should have the following in a field (name is immaterial).
  - It must be in the dataframe that will be updated (even with visibility off)

Live Example - DDP

- MANDATORY
  - Name Field – Title of the map
  - Page Number/Sort field – self evident but think of how the series will look so the numbers should be N-S, W-E or in some defined order.
- OPTIONAL
  - Rotation – if the map is along a river line (for ex)
  - Spatial Ref – only if it changes in certain pages. I
    code can be used.
  - customize->toolbars->data driven pages



- SETUP
  - Extent Tab
    - never use option 1 as the result is not predictable.
    - if all the files will have same scale use option 2
  - if sheets have different scales then the Index should have the scales defined in each row/record. Use just “50000” for 1:50,000 for ex.
    - Careful with this as it will cause scale bars and text to shift. For ex a predefined scale bar that shows 1km at 1:10,000 might show 1.02lm @ 1:12,000 etc..I usually create new mxd for any different scale maps.
- If the file used as the index is updated click refresh on the toolbar.
- Add the custom name and page number.

Live Example - DDP

- You can scroll through pages and
  - use file □ print□ and now the ddp is active.
    - I have found this to be unstable and sometimes crashes when there are 20+ pages to print.

  So I usually use
  - file□export□ pages□ and ‘single pdf’ in the last drop down if there are a lot of pages and you get a lot of quality settings here including ability to convert vector to raster.

Interesting Points - DDP

- centered text blocks may shift unless the automated text is in the same block.
- you can create dynamic text with any attribute field of the Data Driven Pages index layer. For example, if you have an attribute on the index layer named POPULATION, you can create a dynamic text tag using the field <dyn type="page" property="POPULATION"/>
- All items that change must be in the same shape file that is used as the index layer.
- You can use Page Definition Queries to specify how data is drawn on different page views.

Python (Simplified for our workflows)

- No Command Line window as it is replaced by Geoprocessing□Python
- Automate any repetative task
- ARCGIS tools are written in python and are accessible from the arcpy module in python
- The help of any tool will have python code snippets on how to run that command
- Python is used/useful in most software and can be used directly on any OS
- They are all located in Q:\Software\GIS\ESRI\ArcGIS10\CustomScripts
    - Please read the IMPORTANT-Readme.rtf

Python - Continued

- Demo
  - CreateFileList_Ver1a.py
  - DataFrameExtents_toSHP_ver1b.py
  - Raster-Extents.py was used to create the full orthophoto extents file for ex.
- Created tools from some scripts to allow access to them just like any other tool
  - Always check the code to see where any outputs are and how the code will affect your system.
- The toolbox to add/update is at
  Q:\Software\GIS\ESRI\ArcGIS10\CustomScripts\ATGIS.tbx

Junk

- Misc Notes